

# Katravath Srinu

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## EDUCATION

**Indian Institute of Technology (BHU), Varanasi** Varanasi, India  
*B.Tech in Computer Science and Engineering* *Expected May 2027*

- Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, AI, Machine Learning, Generative AI, Probability & Statistics, Distributed Systems
- Secured AIR 11,227 in JEE Advanced 2023 among 150,000+ candidates

## TECHNICAL SKILLS

**Languages:** C++, Python, JavaScript, TypeScript, SQL, HTML, CSS  
**Frameworks & Libraries:** Streamlit, LangChain, Scikit-learn, Node.js, Express.js, Pillow  
**AI / ML:** LLMs, NLP, RAG, Prompt Engineering, Local Model Inference, AI Image Generation  
**Backend & APIs:** REST APIs, WebSockets, Concurrent Session Management, Modular API Architecture  
**Cloud & DevOps:** AWS (EC2, Lambda, API Gateway, S3), Docker, CI/CD Pipelines, Linux  
**Databases:** MongoDB, MySQL, Pinecone (Vector DB)  
**Core CS:** Concurrency & Multi-threading, Distributed Systems, Object-Oriented Design, Algorithm Design & Complexity Analysis  
**Tools:** Git, Postman, Wireshark, Agile/Scrum

## EXPERIENCE

**Independent Open Source Contributor** Remote  
*Backend Software Engineer — Multiple Repositories* *Jan 2025 – Mar 2026*

- Contributed 150+ pull requests improving backend functionality, API performance, and code quality across production repositories; collaborated with distributed teams via Git workflows including feature branching, code review, and issue tracking.
- Implemented concurrent request handling and thread-safe data access patterns; optimized REST API response logic to reduce latency, improve throughput, and prevent regressions under high-load conditions.

## PROJECTS

**Enterprise AI Platform** | *Python, Streamlit, LangChain, LLMs, AWS EC2, REST APIs* 2026

- Architected and deployed a production-grade multi-module AI platform on AWS EC2 with an LLM-powered chat assistant, AI image generation, analytics dashboard, and a prompt-to-website builder generating sites in under 10 minutes.
- Designed modular architecture with concurrent session management, multi-threaded request processing, and a decoupled REST API layer; applied complexity analysis to optimize latency and throughput across simultaneous users.

**VoIP Intrusion Detection System** | *Python, SIP, PCAP, Wireshark, Scikit-learn* 2025

- Built a real-time network security pipeline achieving 94% attack classification accuracy across SPIT, REGISTER flooding, and SIP spoofing; analyzed 10+ GB of PCAP traces to identify adversarial patterns.
- Engineered packet-level feature extraction reducing false positives by 40%; applied  $O(n \log n)$  algorithm optimization for efficient detection on high-throughput concurrent network data streams.

**Real-Time Chat Application** | *Node.js, Express.js, WebSockets, MongoDB* 2024

- Built a scalable multi-user messaging platform with WebSocket-based concurrent connection handling and multi-threaded event processing for low-latency real-time communication across thousands of simultaneous users.
- Implemented RESTful authentication with persistent MongoDB storage; architected for horizontal scalability on distributed infrastructure to support enterprise-level concurrent loads.

## ACHIEVEMENTS & CERTIFICATIONS

- AWS Cloud Practitioner Essentials** — AWS Training & Certification Completion Certificate May 2026
- HackerRank** Software Engineer Intern Certification — Data Structures, Algorithms, SQL, Python Jan 2026
- Solved **300+ DSA problems** on LeetCode & Codeforces; Codeforces Max Rating: **1350**
- Active open-source contributor with **150+ merged pull requests** across multiple production repositories